

Colt Melton

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EDUCATION

Kettering University | Flint, MI

July 2023 – Present

Bachelor of Science, Computer Science

Ajou University | Suwon, South Korea

February 2025 – July 2025

Exchange Program in Software Engineering

TECHNICAL SKILLS

Languages: C#, Python, SQL, C++, Java, JavaScript, C

Software & Development: C#/.NET, Python, REST APIs, OPC UA, VMware Automation, GUI Development, Git, Bamboo

Infrastructure & Cloud: AWS, Linux, Networking, System Administration

PROFESSIONAL EXPERIENCE

Research & Development Engineer / Scientist

October 2025 – October 2026

Endress+Hauser | Ann Arbor, MI

- Developed a real-time spectroscopy monitoring platform integrating **OPC UA telemetry, spectral visualization, and analyzer diagnostics** across **30+ industrial analyzers** for internal and customer-facing use.
- Automated software deployment and validation workflows using **Bamboo, VMware, and Robot Framework**, reducing manual testing effort by approximately **4 hours per release cycle** across multiple analyzer configurations.
- Partnered with lead spectroscopy scientists to develop software supporting **Raman analyzer validation, diagnostics, analytical method development, and customer support investigations**.

Automation Software and Calibration Engineer

October 2023 – July 2025

Bosch International Engineering | Farmington Hills, MI

- Developed automation software utilizing proprietary engineering APIs to streamline analysis of ASCET-generated control software, enabling **calibration engineers** to trace parameter dependencies and investigate system behavior more efficiently.
- Analyzed CAN bus telemetry using **CANoe, CANalyzer, and ETAS tooling** to support calibration, validation, and performance investigations of prototype and production braking systems.
- Investigated ECU intervention behavior through vehicle testing and data analysis, identifying interactions between calibration parameters, control logic, and vehicle dynamics.
- Collaborated with controls, software, and vehicle dynamics engineers to validate advanced braking, chassis, and safety-critical vehicle systems.

RESEARCH EXPERIENCE

Thesis: Machine Learning, Neural Network Raman Spectral Decomposition

- Designed and evaluated a **2.9M-parameter PyTorch 1D U-Net** for Raman spectral decomposition, separating baseline and overlapping peak components across **602 spectra** with **0.9999 mean R² reconstruction accuracy**.
- Developed an end-to-end analysis pipeline using **PyTorch, SciPy, lmfit, and statsmodels** to detect, isolate, fit, and quantify Raman spectral peaks.
- Scaled analysis across **19,753 validated peak matches**, supporting automated decomposition of complex multi-component spectra.
- Built spectrum-level calibration models achieving **R² up to 0.98** for ethane and propane concentration prediction.

PROJECTS

VMware Automation Platform

- Engineered a C#/.NET virtualization automation platform integrating VMware REST APIs, reducing manual test-environment setup through automated VM provisioning, configuration deployment, and validation workflows.
- Implemented JSON-driven configuration management, automated software installation, Robot Framework execution, and VM lifecycle orchestration for engineering test systems.

SCIF SIEM

- Developed a Python Security Information and Event Management platform utilizing custom detection rules, event correlation, and MITRE ATT&CK mapping.

OPC UA Spectroscopy Dashboard

- Developed a real-time industrial telemetry platform using Python and OPC UA, aggregating live spectroscopy measurements and analyzer health metrics to support operational monitoring and predictive diagnostics.

LEADERSHIP & ACTIVITIES

- **Kagle Leadership Initiative(KLI)** - Mentor role for support and guidance for leadership and academic success amongst early college and high school students.
- **President, Cliff Hangers Rock Climbing Club** - Scheduling, planning, event management, budget.

CERTIFICATIONS

- **CompTIA IT Fundamentals (ITF+)**